



MATH230: Gradient and Directional Derivatives

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Computing gradient components

1. For $f(x,y) = 1x^2y^2$, compute $f_x(1,1)$ — the x-component of the gradient at (1,1).

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (1, 1)$$

Answer: _____

2. For $f(x,y) = 1x^2y^2$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

3. For $f(x,y) = 3x^3y^2$, compute $f_x(2,1)$ — the x-component of the gradient at (2,1).

$$f(x,y) = 3x^3y^2, (x_0, y_0) = (2, 1)$$

Answer: _____

4. For $f(x,y) = 1x^2y^2$, compute $f_y(3,3)$ — the y-component of the gradient at (3,3).

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (3, 3)$$

Answer: _____

5. For $f(x,y) = 3x^2y^2$, compute $f_x(3,1)$ — the x-component of the gradient at (3,1).

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (3, 1)$$

Answer: _____

6. For $f(x,y) = 1x^2y^3$, compute $f_y(1,2)$ — the y-component of the gradient at (1,2).

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (1, 2)$$

Answer: _____



7. For $f(x,y) = 2x^2y^1$, compute $f_x(1,3)$ — the x-component of the gradient at (1,3).

$$f(x,y) = 2x^2y^1, (x_0, y_0) = (1, 3)$$

Answer: _____

8. For $f(x,y) = 3x^1y^3$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 3x^1y^3, (x_0, y_0) = (2, 2)$$

Answer: _____

9. For $f(x,y) = 2x^3y^2$, compute $f_x(2,2)$ — the x-component of the gradient at (2,2).

$$f(x,y) = 2x^3y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

10. For $f(x,y) = 1x^1y^2$, compute $f_y(3,2)$ — the y-component of the gradient at (3,2).

$$f(x,y) = 1x^1y^2, (x_0, y_0) = (3, 2)$$

Answer: _____

11. For $f(x,y) = 1x^3y^1$, compute $f_x(1,3)$ — the x-component of the gradient at (1,3).

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (1, 3)$$

Answer: _____

12. For $f(x,y) = 2x^1y^2$, compute $f_y(1,1)$ — the y-component of the gradient at (1,1).

$$f(x,y) = 2x^1y^2, (x_0, y_0) = (1, 1)$$

Answer: _____

13. For $f(x,y) = 3x^3y^1$, compute $f_x(2,1)$ — the x-component of the gradient at (2,1).

$$f(x,y) = 3x^3y^1, (x_0, y_0) = (2, 1)$$

Answer: _____

14. For $f(x,y) = 4x^1y^2$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 4x^1y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

15. For $f(x,y) = 4x^3y^2$, compute $f_x(2,2)$ — the x-component of the gradient at (2,2).

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (2, 2)$$

Answer: _____



16. For $f(x,y) = 4x^1y^3$, compute $f_y(3,3)$ — the y-component of the gradient at (3,3).

$$f(x,y) = 4x^1y^3, (x_0, y_0) = (3, 3)$$

Answer: _____

17. For $f(x,y) = 1x^3y^2$, compute $f_x(1,2)$ — the x-component of the gradient at (1,2).

$$f(x,y) = 1x^3y^2, (x_0, y_0) = (1, 2)$$

Answer: _____

18. For $f(x,y) = 1x^1y^3$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 1x^1y^3, (x_0, y_0) = (2, 2)$$

Answer: _____

19. For $f(x,y) = 1x^2y^1$, compute $f_x(1,3)$ — the x-component of the gradient at (1,3).

$$f(x,y) = 1x^2y^1, (x_0, y_0) = (1, 3)$$

Answer: _____

20. For $f(x,y) = 1x^2y^3$, compute $f_y(2,1)$ — the y-component of the gradient at (2,1).

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (2, 1)$$

Answer: _____

Dot product in gradient context

21. The gradient ∇f at a point is (1, 3, 0) and the unit direction vector is (-2, 0, 0). Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(1, 3, 0) \cdot (-2, 0, 0)$$

Answer: _____

22. The gradient ∇f at a point is (0, -4, 0) and the unit direction vector is (1, 0, 0). Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(0, -4, 0) \cdot (1, 0, 0)$$

Answer: _____

23. The gradient ∇f at a point is (2, -1, 0) and the unit direction vector is (2, -2, 0). Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(2, -1, 0) \cdot (2, -2, 0)$$

Answer: _____



24. The gradient ∇f at a point is $(4, -2, 0)$ and the unit direction vector is $(1, 2, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(4, -2, 0) \cdot (1, 2, 0)$$

Answer: _____

25. The gradient ∇f at a point is $(3, -5, 0)$ and the unit direction vector is $(2, 1, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(3, -5, 0) \cdot (2, 1, 0)$$

Answer: _____

26. The gradient ∇f at a point is $(-2, -4, 0)$ and the unit direction vector is $(-3, -2, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(-2, -4, 0) \cdot (-3, -2, 0)$$

Answer: _____

27. The gradient ∇f at a point is $(4, -3, 0)$ and the unit direction vector is $(3, -2, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(4, -3, 0) \cdot (3, -2, 0)$$

Answer: _____

28. The gradient ∇f at a point is $(1, -1, 0)$ and the unit direction vector is $(1, 3, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(1, -1, 0) \cdot (1, 3, 0)$$

Answer: _____

29. The gradient ∇f at a point is $(2, 3, 0)$ and the unit direction vector is $(1, 1, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(2, 3, 0) \cdot (1, 1, 0)$$

Answer: _____

30. The gradient ∇f at a point is $(0, 3, 0)$ and the unit direction vector is $(-1, -2, 0)$. Compute the directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$.

$$(0, 3, 0) \cdot (-1, -2, 0)$$

Answer: _____





Topics: Computing gradient components, Dot product in gradient context. All answers verified by independent computation.

Solutions



Computing gradient components

1. For $f(x,y) = 1x^2y^2$, compute $f_x(1,1)$ — the x-component of the gradient at $(1,1)$.

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (1, 1)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 2x^1y^2$.

→ $f_x(1,1) = 2$.

→ This is the x-component of ∇f at $(1,1)$.

Answer: $f_x = 2x^1y^2, f_x(1, 1) = 2$

2. For $f(x,y) = 1x^2y^2$, compute $f_y(2,2)$ — the y-component of the gradient at $(2,2)$.

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 2x^2y^1$.

→ $f_y(2,2) = 16$.

→ This is the y-component of ∇f at $(2,2)$.

Answer: $f_y = 2x^2y^1, f_y(2, 2) = 16$

3. For $f(x,y) = 3x^3y^2$, compute $f_x(2,1)$ — the x-component of the gradient at $(2,1)$.

$$f(x,y) = 3x^3y^2, (x_0, y_0) = (2, 1)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 9x^2y^2$.

→ $f_x(2,1) = 36$.

→ This is the x-component of ∇f at $(2,1)$.

Answer: $f_x = 9x^2y^2, f_x(2, 1) = 36$

4. For $f(x,y) = 1x^2y^2$, compute $f_y(3,3)$ — the y-component of the gradient at $(3,3)$.

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (3, 3)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 2x^2y^1$.

→ $f_y(3,3) = 54$.

→ This is the y-component of ∇f at $(3,3)$.

Answer: $f_y = 2x^2y^1, f_y(3, 3) = 54$

5. For $f(x,y) = 3x^2y^2$, compute $f_x(3,1)$ — the x-component of the gradient at $(3,1)$.

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (3, 1)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 6x^1y^2$.

→ $f_x(3,1) = 18$.

→ This is the x-component of ∇f at $(3,1)$.

Answer: $f_x = 6x^1y^2, f_x(3, 1) = 18$



6. For $f(x,y) = 1x^2y^3$, compute $f_y(1,2)$ — the y-component of the gradient at $(1,2)$.

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (1, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 3x^2y^2$.

→ $f_y(1,2) = 12$.

→ This is the y-component of ∇f at $(1,2)$.

Answer: $f_y = 3x^2y^2, f_y(1, 2) = 12$

7. For $f(x,y) = 2x^2y^1$, compute $f_x(1,3)$ — the x-component of the gradient at $(1,3)$.

$$f(x,y) = 2x^2y^1, (x_0, y_0) = (1, 3)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 4x^1y^1$.

→ $f_x(1,3) = 12$.

→ This is the x-component of ∇f at $(1,3)$.

Answer: $f_x = 4x^1y^1, f_x(1, 3) = 12$

8. For $f(x,y) = 3x^1y^3$, compute $f_y(2,2)$ — the y-component of the gradient at $(2,2)$.

$$f(x,y) = 3x^1y^3, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 9x^1y^2$.

→ $f_y(2,2) = 72$.

→ This is the y-component of ∇f at $(2,2)$.

Answer: $f_y = 9x^1y^2, f_y(2, 2) = 72$

9. For $f(x,y) = 2x^3y^2$, compute $f_x(2,2)$ — the x-component of the gradient at $(2,2)$.

$$f(x,y) = 2x^3y^2, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 6x^2y^2$.

→ $f_x(2,2) = 96$.

→ This is the x-component of ∇f at $(2,2)$.

Answer: $f_x = 6x^2y^2, f_x(2, 2) = 96$

10. For $f(x,y) = 1x^1y^2$, compute $f_y(3,2)$ — the y-component of the gradient at $(3,2)$.

$$f(x,y) = 1x^1y^2, (x_0, y_0) = (3, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 2x^1y^1$.

→ $f_y(3,2) = 12$.

→ This is the y-component of ∇f at $(3,2)$.

Answer: $f_y = 2x^1y^1, f_y(3, 2) = 12$



11. For $f(x,y) = 1x^3y^1$, compute $f_x(1,3)$ — the x-component of the gradient at (1,3).

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (1, 3)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 3x^2y^1$.

→ $f_x(1,3) = 9$.

→ This is the x-component of ∇f at (1,3).

Answer: $f_x = 3x^2y^1, f_x(1, 3) = 9$

12. For $f(x,y) = 2x^1y^2$, compute $f_y(1,1)$ — the y-component of the gradient at (1,1).

$$f(x,y) = 2x^1y^2, (x_0, y_0) = (1, 1)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 4x^1y^1$.

→ $f_y(1,1) = 4$.

→ This is the y-component of ∇f at (1,1).

Answer: $f_y = 4x^1y^1, f_y(1, 1) = 4$

13. For $f(x,y) = 3x^3y^1$, compute $f_x(2,1)$ — the x-component of the gradient at (2,1).

$$f(x,y) = 3x^3y^1, (x_0, y_0) = (2, 1)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 9x^2y^1$.

→ $f_x(2,1) = 36$.

→ This is the x-component of ∇f at (2,1).

Answer: $f_x = 9x^2y^1, f_x(2, 1) = 36$

14. For $f(x,y) = 4x^1y^2$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 4x^1y^2, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 8x^1y^1$.

→ $f_y(2,2) = 32$.

→ This is the y-component of ∇f at (2,2).

Answer: $f_y = 8x^1y^1, f_y(2, 2) = 32$

15. For $f(x,y) = 4x^3y^2$, compute $f_x(2,2)$ — the x-component of the gradient at (2,2).

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 12x^2y^2$.

→ $f_x(2,2) = 192$.

→ This is the x-component of ∇f at (2,2).

Answer: $f_x = 12x^2y^2, f_x(2, 2) = 192$



16. For $f(x,y) = 4x^1y^3$, compute $f_y(3,3)$ — the y-component of the gradient at (3,3).

$$f(x,y) = 4x^1y^3, (x_0, y_0) = (3, 3)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 12x^1y^2$.

→ $f_y(3,3) = 324$.

→ This is the y-component of ∇f at (3,3).

Answer: $f_y = 12x^1y^2, f_y(3, 3) = 324$

17. For $f(x,y) = 1x^3y^2$, compute $f_x(1,2)$ — the x-component of the gradient at (1,2).

$$f(x,y) = 1x^3y^2, (x_0, y_0) = (1, 2)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 3x^2y^2$.

→ $f_x(1,2) = 12$.

→ This is the x-component of ∇f at (1,2).

Answer: $f_x = 3x^2y^2, f_x(1, 2) = 12$

18. For $f(x,y) = 1x^1y^3$, compute $f_y(2,2)$ — the y-component of the gradient at (2,2).

$$f(x,y) = 1x^1y^3, (x_0, y_0) = (2, 2)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 3x^1y^2$.

→ $f_y(2,2) = 24$.

→ This is the y-component of ∇f at (2,2).

Answer: $f_y = 3x^1y^2, f_y(2, 2) = 24$

19. For $f(x,y) = 1x^2y^1$, compute $f_x(1,3)$ — the x-component of the gradient at (1,3).

$$f(x,y) = 1x^2y^1, (x_0, y_0) = (1, 3)$$

→ Gradient = (f_x, f_y) . Find x-component: $f_x = 2x^1y^1$.

→ $f_x(1,3) = 6$.

→ This is the x-component of ∇f at (1,3).

Answer: $f_x = 2x^1y^1, f_x(1, 3) = 6$

20. For $f(x,y) = 1x^2y^3$, compute $f_y(2,1)$ — the y-component of the gradient at (2,1).

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (2, 1)$$

→ Gradient = (f_x, f_y) . Find y-component: $f_y = 3x^2y^2$.

→ $f_y(2,1) = 12$.

→ This is the y-component of ∇f at (2,1).

Answer: $f_y = 3x^2y^2, f_y(2, 1) = 12$



Dot product in gradient context

21. The gradient ∇f at a point is $(1, 3, 0)$ and the unit direction vector is $(-2, 0, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(1, 3, 0) \cdot (-2, 0, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (1)(-2) + (3)(0) + 0.$$

$$\rightarrow = -2.$$

Answer: $= -2$

22. The gradient ∇f at a point is $(0, -4, 0)$ and the unit direction vector is $(1, 0, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(0, -4, 0) \cdot (1, 0, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (0)(1) + (-4)(0) + 0.$$

$$\rightarrow = 0.$$

Answer: $= 0$

23. The gradient ∇f at a point is $(2, -1, 0)$ and the unit direction vector is $(2, -2, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(2, -1, 0) \cdot (2, -2, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (2)(2) + (-1)(-2) + 0.$$

$$\rightarrow = 6.$$

Answer: $= 6$

24. The gradient ∇f at a point is $(4, -2, 0)$ and the unit direction vector is $(1, 2, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(4, -2, 0) \cdot (1, 2, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (4)(1) + (-2)(2) + 0.$$

$$\rightarrow = 0.$$

Answer: $= 0$

25. The gradient ∇f at a point is $(3, -5, 0)$ and the unit direction vector is $(2, 1, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(3, -5, 0) \cdot (2, 1, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (3)(2) + (-5)(1) + 0.$$

$$\rightarrow = 1.$$

Answer: $= 1$

26. The gradient ∇f at a point is $(-2, -4, 0)$ and the unit direction vector is $(-3, -2, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(-2, -4, 0) \cdot (-3, -2, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (-2)(-3) + (-4)(-2) + 0.$$

$$\rightarrow = 14.$$

Answer: $= 14$



27. The gradient ∇f at a point is $(4, -3, 0)$ and the unit direction vector is $(3, -2, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(4, -3, 0) \cdot (3, -2, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (4)(3) + (-3)(-2) + 0.$$

$$\rightarrow = 18.$$

Answer: $= 18$

28. The gradient ∇f at a point is $(1, -1, 0)$ and the unit direction vector is $(1, 3, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(1, -1, 0) \cdot (1, 3, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (1)(1) + (-1)(3) + 0.$$

$$\rightarrow = -2.$$

Answer: $= -2$

29. The gradient ∇f at a point is $(2, 3, 0)$ and the unit direction vector is $(1, 1, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(2, 3, 0) \cdot (1, 1, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (2)(1) + (3)(1) + 0.$$

$$\rightarrow = 5.$$

Answer: $= 5$

30. The gradient ∇f at a point is $(0, 3, 0)$ and the unit direction vector is $(-1, -2, 0)$. Compute the directional derivative $D_u f = \nabla f \cdot u$.

$$(0, 3, 0) \cdot (-1, -2, 0)$$

$$\rightarrow D_u f = \nabla f \cdot u = (0)(-1) + (3)(-2) + 0.$$

$$\rightarrow = -6.$$

Answer: $= -6$

